Deep Learning Temporal Hierarchies for Interval Forecasts

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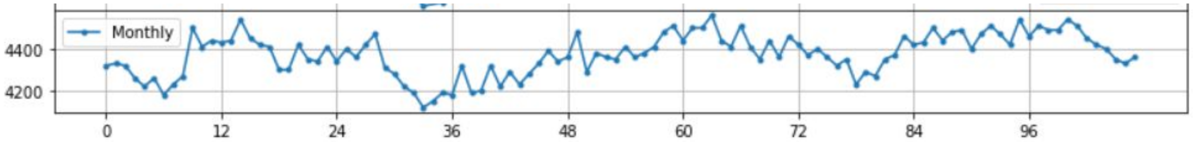
Temporal Hierarchies … in a Nutshell



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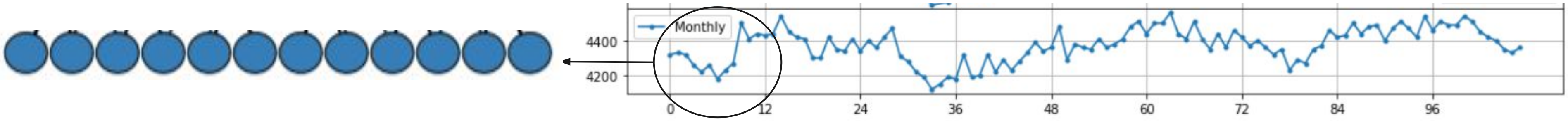

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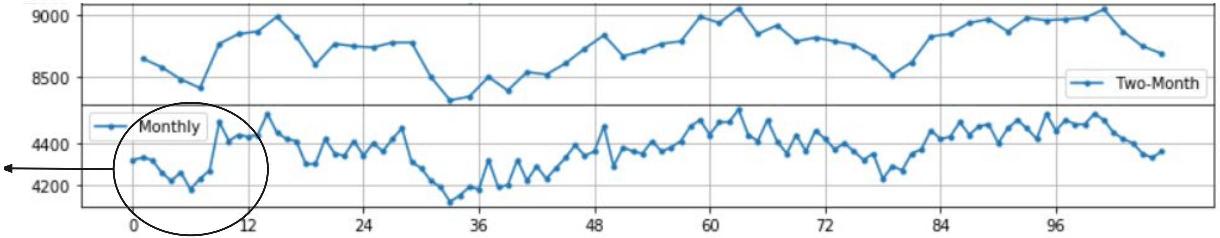

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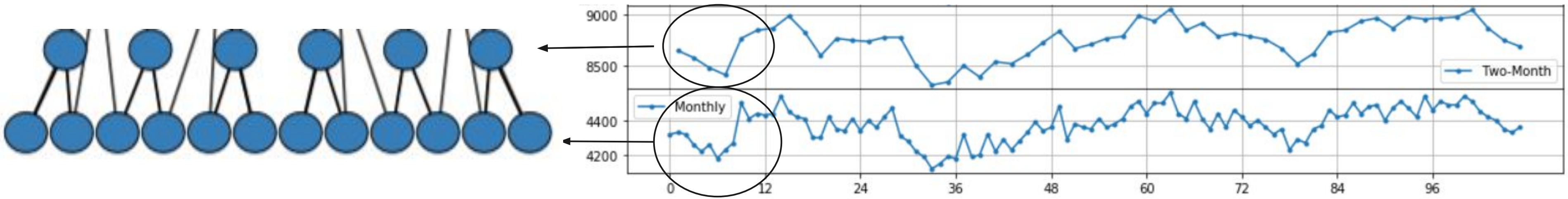

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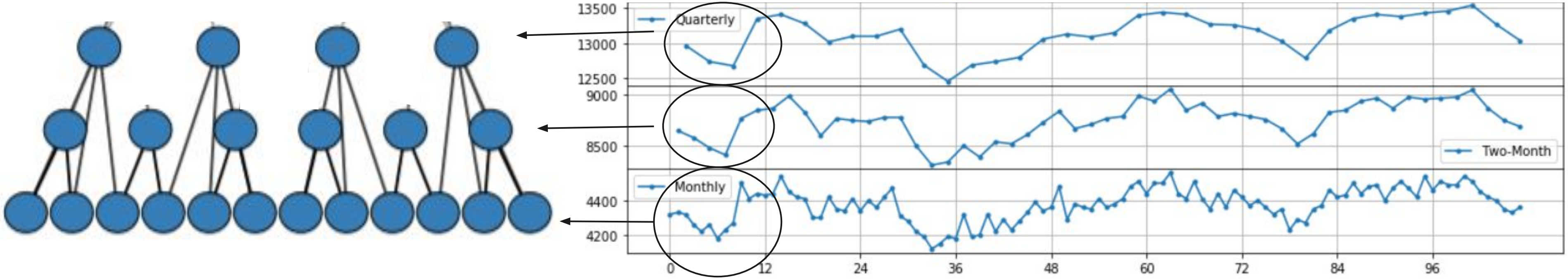

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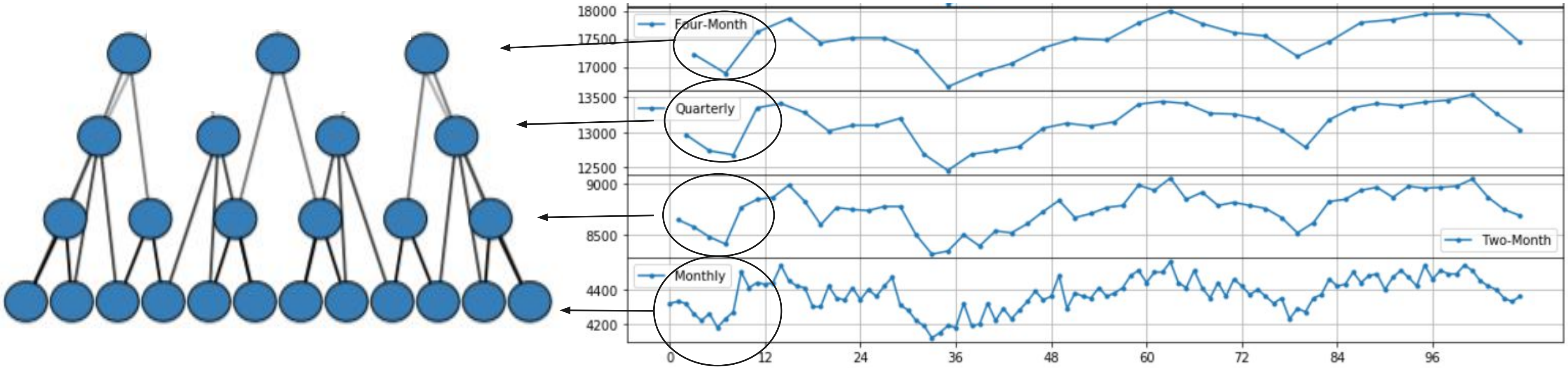

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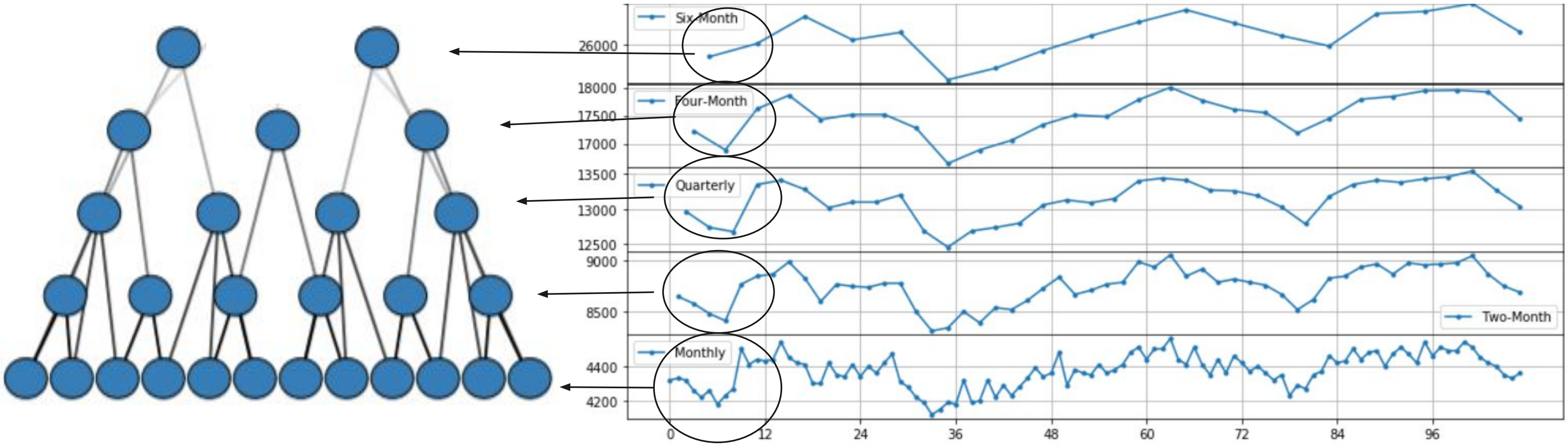

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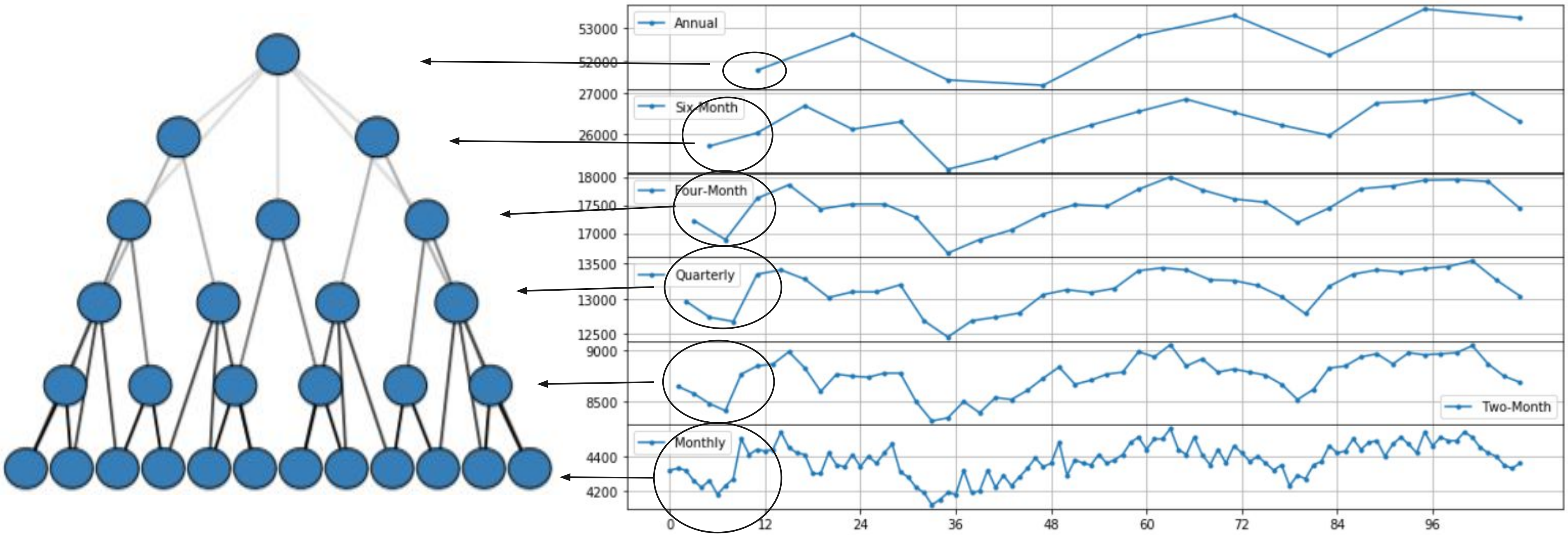

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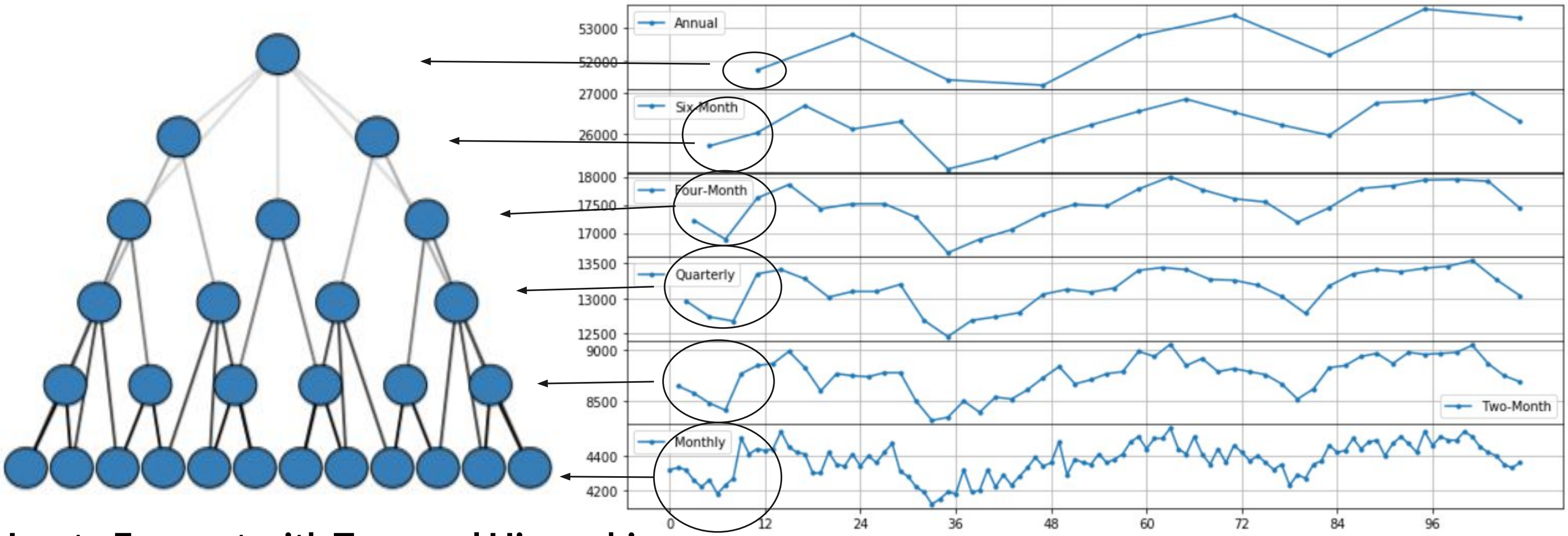

How to Forecast with Temporal Hierarchies

- Bottom - Up, Top - Down
- Base Forecasts & Reconciliation
- Simultaneous Forecasts

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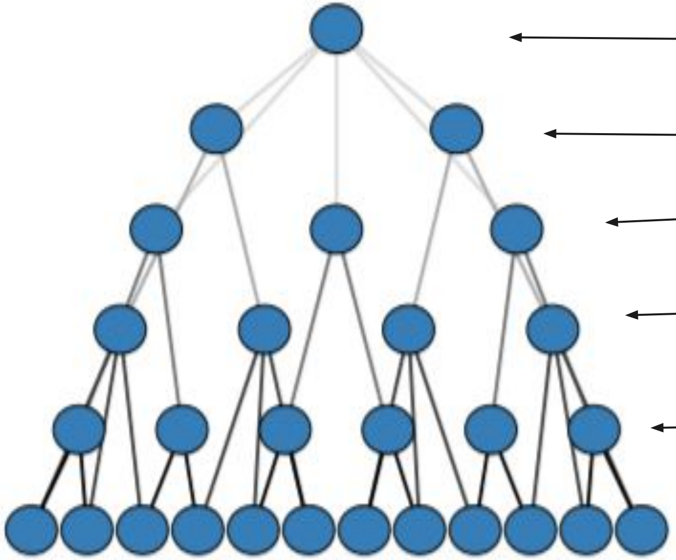

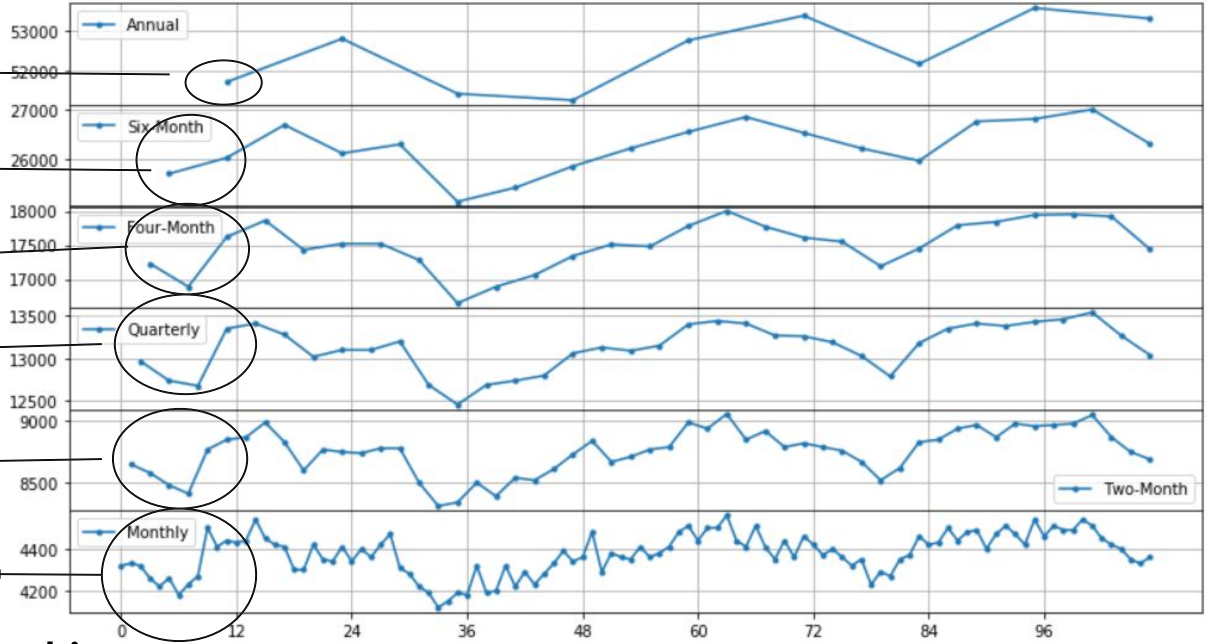

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Empirical Evidence pointing out Temporal Hierarchies have accuracy benefit over base forecasts

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Temporal Hierarchies in Finance: Challenges & Goal

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Two key challenges in stock predictions:

1. Pick the right granularity and forecast horizon
   a. Short/Mid/Long term forecasts.
   b. Restricted strategy planning
   c. Multiple models for holistic modelling
2. Handle Uncertainty
   a. Inherited uncertainty of stock market
   b. Focus on point (mean) forecast → Low signal to noise ratio

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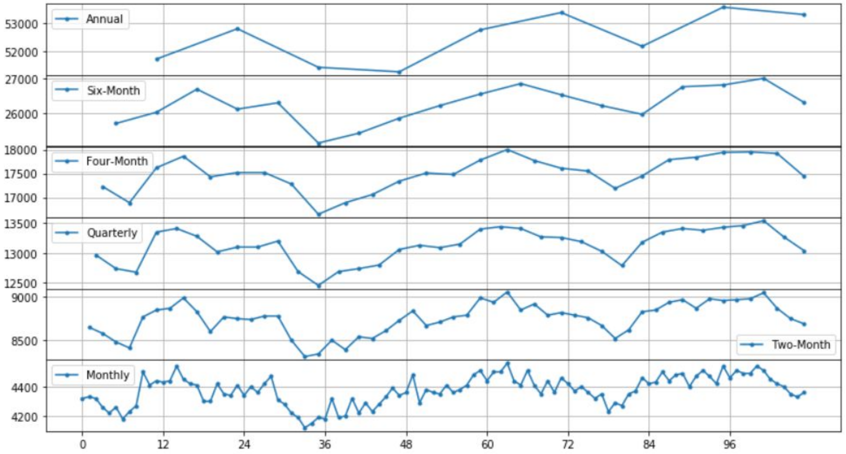
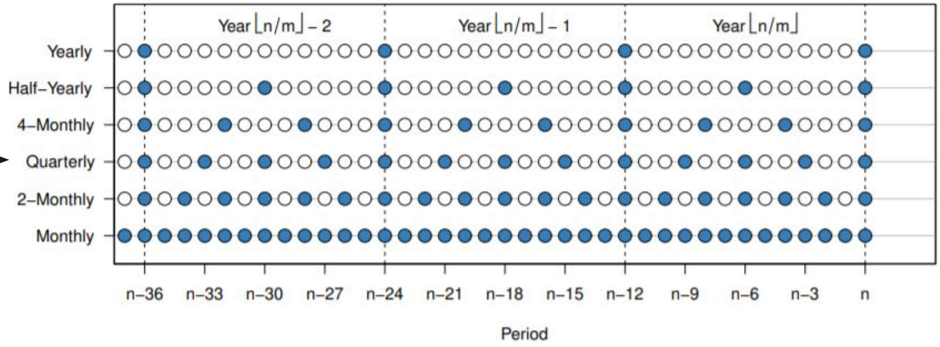
Goal of this work:

- Merge temporal hierarchies & deep learning for prediction interval forecasting
- Simultaneous multi-horizon forecasts
- Promising empirical alternatives

DeepTHieF: Base Forecast & Reconciliation in a go.





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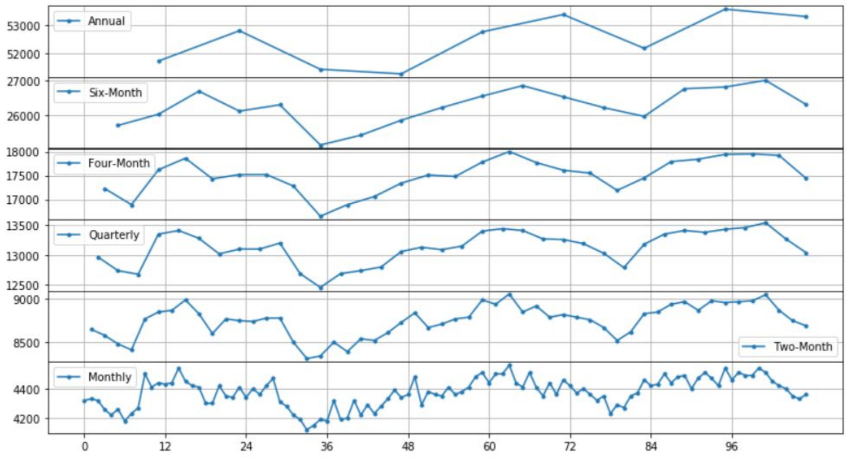

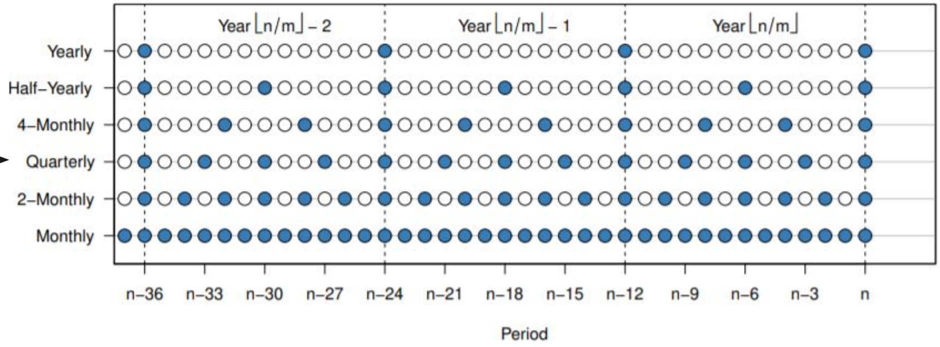

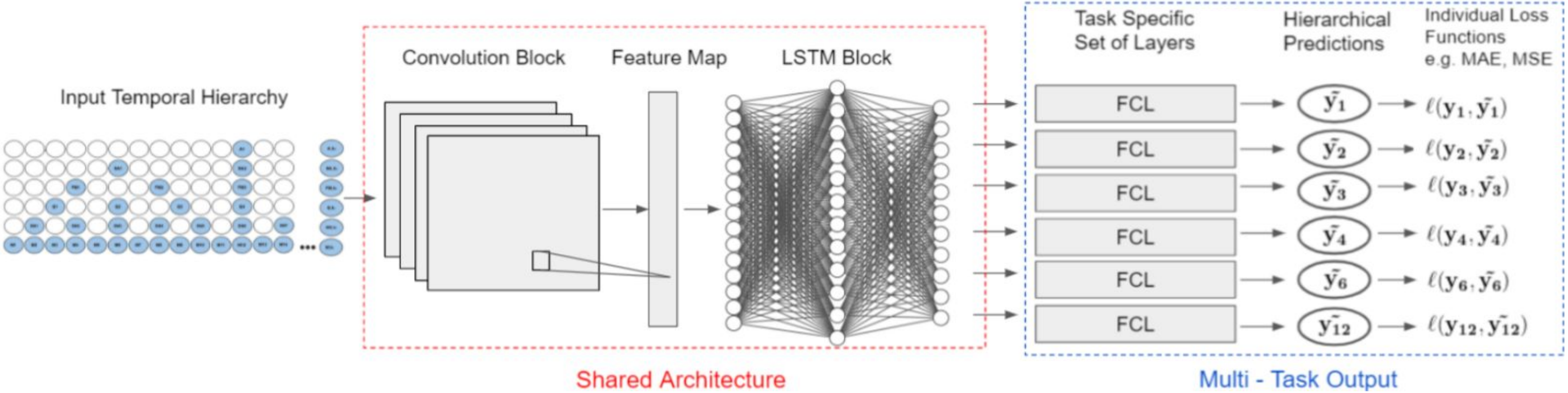

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Methods for Estimating Prediction Intervals

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- Direct Optimization
  - Modify output layers
  - Output lower and upper quantile for each hierarchical level
  - Interval Score as a loss function $(u-l)+\frac{2}{a}(l-x)*ID(x,l)+\frac{2}{a}(x-u)ID(u,x)$

- Empirical Methods: Assume future errors have similar behaviour to past ones
  - Overlap strong assumptions
  - Extract the upper and lower quantiles from the in-sample error distribution
    - Kernel Density Estimator
    - Direct extraction
  - Produce a point forecast and sum the upper and lower quantiles

$$PI = [MeanForecast + LowInterval, MeanForecast + UpperInterval]$$

<u>DeepTHieF is used a forecasting model to simultaneously produce intervals for multiple levels</u>

Experimental Design

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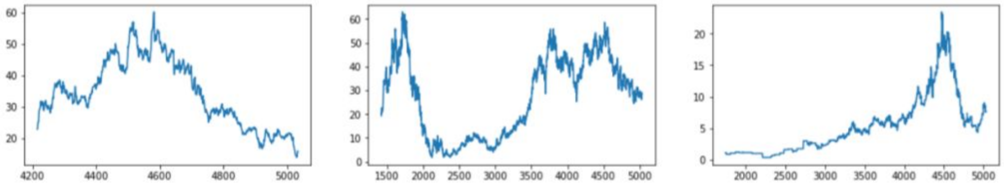
The Aim: Forecast one-step-ahead prediction intervals for log difference of stocks indices.

- Data:
  - 4000 daily stocks found in NASDAQ
  - A full quarter (98 weeks) kept as test set
- Fitting DeepTHieF:
  - Data Augmentations
  - Fitting period composed of 365 days
  - Temporal aggregations for factors of 7, 91, 364 (days in week, quarter, year)
- Evaluation Metrics
  - Daily - Weekly - Monthly - Quarterly Forecasts for holistic strategy
  - Scaled Interval Score
- Benchmarks:
  - ETS (analytical & simulation methods)
  - Naive (analytical)



https://www.kaggle.com/raymondsunartio/6000-nasdaq-stocks-historical-daily-prices

Rob J. Hyndman and George Athanasopoulos. 2018. Forecasting: principles and practice (2nd ed.). OTexts.

Results

Model	Method	Daily	Weekly	Monthly	Quarterly	Mean
DeepTHieF	Empirical - Direct	3.997	3.993	2.774	3.144	3.477
	Empirical - KDE	5.430	5.306	3.589	3.358	4.421
	Interval Optimization	**1.125**	**2.138**	**1.774**	**2.500**	**1.884**
ETS	Analytical	4.023	4.081	4.239	7.034	4.844
	Simulation	4.023	4.082	4.240	7.039	4.846
Naïve	Analytical	5.714	5.747	5.866	9.601	6.731

Table 1: Interval Scores comparison for the proposed methods

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- DeepTHieF results are significantly more accurate
- Interval Optimization performs best with Empirical methods showing competitive performance

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But why??

1. (Globally) Optimized for interval forecasting
2. Captures different structural information at each level
3. Identifies long-term interdependencies in high sampling frequencies
4. Multi-Task learning regularizes shared representations
5. Benefit long-term strategies by modelling cleaner signals on higher-frequency levels.

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To Conclude

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A novel approach to forecast prediction intervals for multiple horizons

- Input representation driven by Temporal Hierarchies
- Simultaneous multi-horizon forecasts enhancing decision making
  (Also assisting in constructing informative shared representations)
- Direct optimization on prediction intervals
  (Empirical methods as a useful alternative)

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Generalizable methodology for any global model.

To do:

Minimize both interval and point forecast loss.

Thank you for your Attention!!

:)

Questions?

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